Cuda lab 3

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1 Part 1 warps

When comparing the full vs partial warps there is a clear difference when it comes to the execution speed. This can be seen in figure 1. At every interval of 32 a slight increase in execution time can be seen, this is because of the sub-optimal core allocation. When partial wraps are used it means that there are some unused GPU threads in the warp. This will then result in less optimal GPU utilisation and thus a slower execution time.

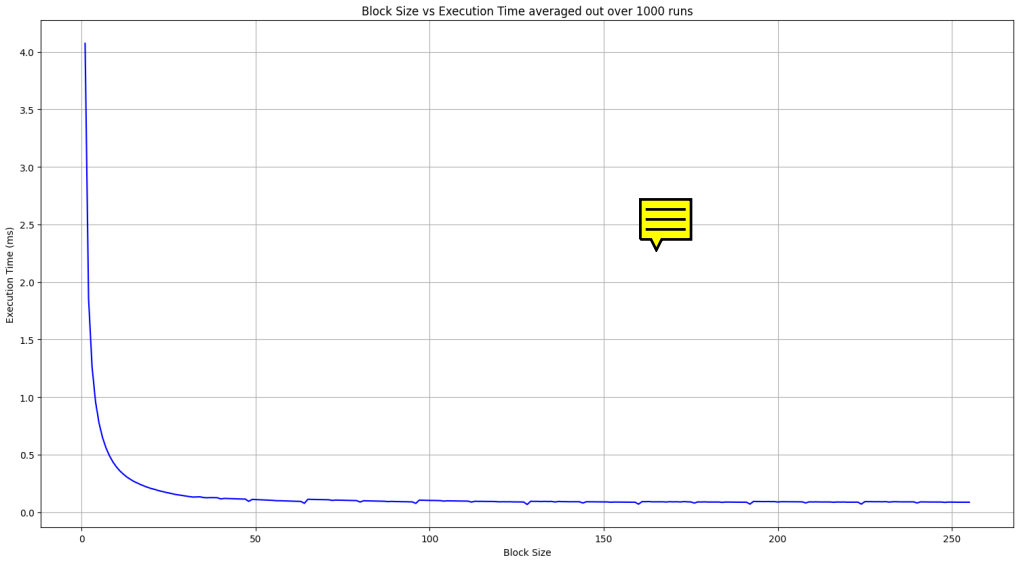


Figure 1: The block size vs the execution time averaged

2 Part 2 thread divergence

For the second part we will make an extra time complex operation on one of the RGB values. This so that one thread will take slightly longer to compute then the others. Since the threads of a CUDA GPU always have to wait for the warp to finish. This will halt all the finished threads which have a faster execution time. Because of this effect it is beneficial to compute all similar operations or at least operation which have a similar execution time in the same warps. This will reduce the total idling time of the different threads, which will speed the whole process up.

With our code implementation the image takes 758 microseconds to compute the image with thread divergence thus running rgbrgbrgb... and 634 microseconds with the thread divergence fix and thus when computing rrrr...gggg...bbbb... to reduce the thread idling time in the warps. Our entire timing results can be found together with our code in our Github Repository.